

Salinity dependence of complex conductivity of unconsolidated and consolidated materials: Comparisons with electrical double layer models

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ABSTRACT

We examined the dependence of imaginary conductivity (σ'') on pore fluid conductivity (σ_w) for an extensive database of 67 samples acquired from twelve independent studies. We compared fitting of functions describing the salinity dependence of σ'' for two models of the electrical double layer (EDL) polarization, both of which predict asymptotic behavior of σ'' at high σ_w . We define these models as the diffuse layer polarization (DLP) and Stern layer polarization (SLP) models based on the physical description of the salinity dependence of the surface polarization. We also examined the database for evidence of a high salinity decrease in σ'' not predicted by either model. The dependence of σ'' on σ_w prior to the polarization plateau predicted by both models approximates a simple empirical power law with an average exponent of 0.34. The salinity dependence predicted by the DLP model adequately describes

most data sets. A fitting parameter representing the high salinity σ'' asymptote is strongly correlated ($R^2 = 0.822$) with pore normalized specific surface (S_{por}). The SLP model describes well the observations when a recently proposed additive polarization term representing the contribution of the protons is included. In this case, the SLP model provides an excellent fit to the data sets, including a low salinity asymptote (in log-log conductivity space) seen in some samples. Predicted values of the fitting parameters of the SLP model generally are consistent with the values expected based on the theory; the fitting parameter describing the high salinity asymptote of the SLP model is also strongly correlated ($R^2 = 0.890$) with S_{por} . The SLP and DLP models neglect a high salinity decrease in the polarization that is observed in numerous data sets from independent studies. New data acquired on a sandstone sample demonstrate that this high salinity decrease is likely not attributable to the limited phase accuracy of earlier measurements.

INTRODUCTION

The induced polarization (IP) method has emerged as a potentially powerful tool for subsurface imaging due to the dependence of the measurements on rock or soil internal surface area. For example, Weller et al. (2010b) showed how 114 samples from 10 independent data sets could be described well by a linear relationship between imaginary conductivity (σ'') and the surface area to pore volume (S_{por}). Earlier evidence of such a relationship (Börner and Schön, 1991; Börner, 1992; Börner et al., 1996; Slater and Lesmes, 2002) inspired efforts to use IP measurements to predict permeability (Börner et al., 1996; Slater and Lesmes, 2002; Revil and Florsch, 2010; Weller et al., 2010a).

As a result of such work, understanding of the dependence of IP measurements on the physical characteristics of rocks has evolved to the point that Weller et al. (2010b) cautiously proposed that the relationship between σ'' and S_{por} essentially represents the IP equivalent of Archie's Law (Archie, 1942), an empirical formulation long used to relate resistivity to porosity and water saturation. However, the contribution of the pore fluid chemical composition to the electrical double layer (EDL) polarization recorded with IP is less well defined, although several researchers have previously studied the controls of pore fluid salinity (Klein and Sill, 1982; Vinegar and Waxman, 1984; Lesmes and Frye, 2001; Revil and Skold, 2011; Weller et al., 2011), in addition to pH (Lesmes and Frye, 2001; Skold et al., 2011), and valence (Börner, 1992; Revil and Skold,

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2011; Weller et al., 2011). A notable advance in the mechanistic understanding of fluid chemistry effects is the work of Vaudelet et al. (2011), who combined a mechanistic model (discussed later) with a surface complexation model to predict the effect of differing sorption properties for the mineral surface for different cations.

From a theoretical standpoint, models have been proposed that include a prediction for the expected salinity dependence of IP measurements. One such early model is based on the polarization of the EDL surrounding the pores making up the porous medium, which is conceptualized as a bunch of capillary tubes (Börner, 1991, 1992). This model, which we will refer to here as the “diffuse layer polarization (DLP) model” because the salinity dependence relates to a varying capacity of the diffuse layer, predicts an approximate dependence of σ'' on the square root of the fluid electrical conductivity (σ_w) at low salinity. More recently, mechanistic models for the polarization of porous media have been based around the polarization of the EDL surrounding the mineral grains making up a porous medium (Leroy et al., 2008; Revil and Florsch, 2010). This model, which we will refer to here as the “Stern layer polarization (SLP) model” assumes that, for low frequencies (typically less than 100 Hz), the main polarization of a granular porous material is due to the Stern layer. The SLP model uses a triple layer model (TLM) of the specific surface conductivity of the Stern layer to define the expected salinity dependence associated with the polarization of absorbed counterions.

The DLP model and SLP models predict an asymptotic high-salinity value for σ'' , having been observed in numerous data sets (Vinegar and Waxman, 1984; Lesmes and Frye, 2001; Slater and Glaser, 2003; Weller et al., 2011). Examples of this behavior will be presented later. However, some previous works have observed an apparent decrease in the polarization at high salinity that is not predicted by either model (Vinegar and Waxman, 1984; Flath, 1989; Lesmes and Frye, 2001; Slater and Glaser, 2003). Some workers have suggested that this polarization decrease may result from a decrease in the mobility of the counterions in the EDL at high-ionic concentrations (Lesmes and Frye, 2001). However, Revil and Skold (2011) note that the high-salinity polarization peak is not consistent between data sets, suggesting that the apparent decrease in polarization at high salinity could be an artifact of the measurements resulting from the limited accuracy in the measurement of very low-phase angles at high salinity (discussed later).

In this study, we seek to resolve some of the uncertainty in the dependence of IP measurements on salinity. Whereas Weller et al. (2011) examined the salinity dependence for three sandstone samples, this paper compiles and examines a wide database comprised of 67 samples that have multiple salinity measurements available. Similar to Weller et al. (2010b), we seek to identify consistent empirical relationships across a broad range of consolidated and unconsolidated sedimentary materials recorded by multiple investigators. Furthermore, we extend the work of Weller et al. (2011) by examining the predictive capabilities of the DLP and SLP models with respect to the salinity dependence of the polarizability, and we also present new data that addresses the question regarding a decrease in polarization at high salinity.

ELECTRICAL PROPERTIES

The complex conductivity (σ^*) of a sample is determined from measurements of the conductivity magnitude ($|\sigma|$) and phase shift (φ) of a sample relative to a reference resistor. The real (σ') and

imaginary (σ'') components of the complex conductivity represent the ohmic conduction and polarization charge transport mechanisms respectively. These electrical properties directly are determined from the measurements. The phase shift is related to the real and imaginary parts of the complex conductivity according to

$$\varphi = \arctan\left(\frac{\sigma''}{\sigma'}\right) \approx \frac{\sigma''}{\sigma'} (\varphi < 100 \text{ mrad}). \quad (1)$$

By convention, the phase is defined as positive in conductivity space.

Most models for the complex electrical conductivity of a porous material at low frequencies (e.g., less than 100 Hz) are based on a parallel addition of two conduction terms representing, (1) an electrolytic contribution via conduction through the interconnected pore space (σ_{el}), and (2) a mineral surface contribution via conduction and polarization within the electrical double layer (EDL) of the interconnected pore surface (σ^*_{surf}) (e.g., Vinegar and Waxman, 1984),

$$\sigma^* = \sigma_{el} + \sigma^*_{surf}. \quad (2)$$

Polarization only is associated with the surface conductivity at low frequencies. In terms of the real and imaginary components,

$$\sigma' = \frac{1}{F} \sigma_w + \sigma'_{surf}, \quad (3)$$

and

$$\sigma'' = \sigma''_{surf}. \quad (4)$$

This paper focuses on how σ'' measurements, which directly represent the surface polarization (equation 4), depend on salinity.

Weller et al. (2010b, 2011) defined the fitting parameter of their linear relationship between σ'' and S_{por} as the specific polarizability (c_p),

$$c_p = \frac{\sigma''}{S_{por}}. \quad (5)$$

This parameter therefore represents the polarizability of the mineral-fluid interface per unit S_{por} . Based on a NaCl saturating solution with σ_w approximately equal to 100 mS/m, Weller et al. (2010b) found $c_p \approx 10^{-11} \text{ S} = 10 \text{ pS}$. In this work, we seek to investigate the dependence of c_p on salinity, i.e., $c_p(\sigma_w)$. The dependence of c_p on σ_w is expected, assuming that changes in surface charge density and/or surface ionic mobility (among other factors) modify the surface polarization. This property bears similarities to the specific surface conductance that has been defined in the development of mechanistic models to describe the surface conductivity of a porous medium (Revil and Florsch, 2010). Although c_p as presented by Weller et al. (2010b) is an empirically derived coefficient directly defining the relation between σ'' and S_{por} , it was recently defined in a mechanistic model for shaly sands (Revil, 2012).

The DLP model

Physics-based descriptions of the surface polarization can offer valuable insights into the expected dependence of σ'' on σ_w . We first consider an approach that uses as a starting point a model describing the modulus of the surface conductivity ($|\sigma_{surf}|$) proposed

by [Rink and Schopper \(1974\)](#). The geometrical and physical basis for this model briefly is stated here. Assuming a capillary bundle model, the measurable surface conductivity of a sample can be described by the ratio of cross section of the EDL (A'') to cross section of the sample (A) multiplied by the intrinsic conductivity within the double layer (σ_x),

$$\left| \sigma_{\text{surf}} \right| = \frac{A''}{A} \sigma_x. \quad (6)$$

Parameter A'' can be determined from the circumference of the pore multiplied by the thickness (δ) of the EDL. Considering a cylindrical cross section of the capillaries with a uniform radius r , we get the area of the cross section of a single pore $A' = \pi r^2$. Recognizing that (1) the ratio A'/A is equal to the porosity (Φ), and (2) $A''/A' = 2\delta/r$ is represented by the specific pore space surface area (S_{por}) in capillary bundle models multiplied by the EDL thickness (δ), equation 6 takes the form:

$$\left| \sigma_{\text{surf}} \right| = \frac{A''}{A'} \frac{A'}{A} \sigma_x = \frac{2}{r} \delta \Phi \sigma_x = S_{\text{por}} \delta \Phi \sigma_x. \quad (7)$$

Regarding the tortuous conduction path through the sample, the tortuosity (T) causes a reduction in surface conductivity. Using the ratio T/Φ as an equivalent for the formation factor F , the modulus of surface conductivity can be formulated as

$$\left| \sigma_{\text{surf}} \right| = \frac{S_{\text{por}}}{F} \delta \sigma_x. \quad (8)$$

[Börner \(1991, 1992\)](#) showed that the real (σ'_{surf}) and imaginary (σ''_{surf}) parts of σ^*_{surf} linearly are related and introduced parameter l to represent the ratio of σ''_{surf} to σ'_{surf} . Therefore, assuming the validity of equation 4,

$$\sigma'' \propto \frac{S_{\text{por}}}{F}. \quad (9)$$

The DLP model conceptualizes the fixed (Stern) and diffuse layers as two capacitors and assumes that the polarization of the EDL can be described by the capacitive contributions from both layers. We stress that the model assumes polarization of the fixed (Stern) and diffuse part of the double layers in response to an electrical field. We refer to this model as the DLP model only because it is the polarization of the diffuse layer that is assumed to be dependent on the salinity, and our work specifically is focused on this salinity dependence. This model implicitly assumes that the secondary electric field associated with the polarization is normal to the pore (or grain) surface. The model predicts

$$\sigma'' = l \frac{S_{\text{por}}}{F} f(\sigma_w) \propto \frac{S_{\text{por}}}{F} f(\sigma_w), \quad (10)$$

where l is considered salinity-independent. This factor is justified by the fact that the surface conductivity and polarizability share the same geometry ([Kulenkampff and Schopper, 1988](#)); the same salinity dependence is therefore assumed for the real and imaginary parts of surface conductivity through the dependence on variation in the thickness of the EDL. [Börner \(1992\)](#) presents supporting experimental evidence, finding that l (his equation 15) is nearly

independent of salinity but slightly different in various types of sandstones. Furthermore, [Börner \(1991, 1992\)](#) experimentally confirms the proportionality of the real and imaginary parts of surface conductivity.

The diffuse layer has a salinity-dependent thickness due to the decrease in thickness of the diffuse layer (δ) with increasing ionic concentration (C_f) as predicted, e.g., by traditional Gouy-Chapman double layer theory,

$$\delta = \left(\frac{\epsilon k T}{8 \pi N Z^2 e^2 C_f} \right)^{0.5}, \quad (11)$$

where ϵ is the mean relative permittivity of the diffuse layer, Z is electrolyte valence, N is Avogadro's number, k is Boltzman's constant, T is absolute temperature (in K) and e is the electronic charge ([Stumm and Morgan, 1996](#)).

The thickness is therefore approximately proportional to $C_f^{-0.5}$, implying an increase in capacity that is approximately proportional to $C_f^{0.5}$. The Stern layer is assumed to have a fixed capacity, which implies an asymptotic value of σ'' for high σ_w . The salinity dependence of σ'' predicted by this model can be described as ([Börner, 1991, 1992](#)),

$$\sigma''(\sigma_w) = a_d \frac{\sqrt{\sigma_w}}{b_d + \sqrt{\sigma_w}}, \quad (12)$$

where a and b are fitting parameters, and subscript d is to signify that these are the fitting parameters for the DLP model. Parameter a_d should be linearly dependent on the ionic mobility of the counterions in the diffuse layer (β_d), which for simplicity is considered to be independent of salinity.

The SLP model

The SLP model has been extensively described in recent literature ([Leroy et al., 2008](#); [Revil and Florsch, 2010](#); [Revil and Skold, 2011](#)) and only briefly is described here, with specific focus on the implications for the dependence of σ'' on salinity, for brevity. This model combines a macroscopic conductivity model for the polarization of a porous medium with a triple layer model for the specific surface conductivity of the Stern layer. Such triple layer models are more sophisticated than the traditional double layer model because they incorporate an inner and outer Stern layer and account for specific adsorption of ions that are tightly bound to the surface in the inner layer.

In the grain model, the surface polarization is specifically associated with the Stern layer surrounding the mineral grains, although surface conduction is assumed to occur in the diffuse and Stern layers. The polarization of the Stern layer around a mineral grain is based on a modification of the model of [Schwarz \(1962\)](#) for colloidal suspensions. Ion migration is tangential and results in accumulation of counterions on the side of the grain in the direction of the electric field. This model predicts,

$$\sigma''(\omega = 1/\tau) = \frac{F - 1}{F} \frac{\sum^s(\sigma_w)}{2d_0}, \quad (13)$$

where d_0 is the grain diameter (when the material is composed of uniform grains) and a decisive parameter describing the

polarization, Σ^s is the specific surface conductivity (in S), and τ (in s) is the main relaxation time. Note that this formulation can be extended for a distribution of grain sizes as described in [Revil and Florsch \(2010\)](#).

Assuming spherical particles, we can write,

$$d_0 = \frac{1 - \Phi}{\Phi} \times \frac{6}{S_{\text{por}}}, \quad (14)$$

with Φ being the porosity.

Therefore, by combining equation 14 with equation 13 and using the approximation

$$\Phi \approx \frac{1}{F}, \quad (15)$$

we get

$$\sigma''(\omega = 1/\tau) \propto \frac{S_{\text{por}}}{F} \Sigma^s(\sigma_w). \quad (16)$$

The specific surface conductivity represents the excess conductivity due to the presence of the Stern layer. The salinity dependence of the imaginary conductivity is attributed to the salinity dependence Σ^s determined by the sorption of counterions. The model assumes that the imaginary conductivity exhibits the same relative dependence on salinity of the pore water as Σ^s , i.e., $\partial \sigma'' / \partial C_f = \partial \Sigma^s / \partial C_f$ ([Revil and Skold, 2011](#)).

[Revil and Skold \(2011\)](#) derive the salinity dependence of Σ^s for a binary symmetric 1:1 electrolyte from first principles based on a triple layer model of the mineral-fluid interface for silica. In the case of Na counterions,

$$\Sigma^s(C_f) = \beta_{\text{Na}} e \Gamma_s^0 \frac{K_{\text{Na}} C_f}{10^{-\text{pH}} + K_{\text{Na}} C_f}, \quad (17)$$

where C_f is in mol l^{-1} , K_{Na} is the sorption coefficient for sodium (dimensionless), Γ_s^0 is total site density at the silica surface, and e is the electronic charge. A generalization to a multicomponent electrolyte is presented in [Revil and Florsch \(2010\)](#).

Assuming a constant pH this model predicts the following salinity dependence,

$$\sigma''(\sigma_w) \propto \Sigma^s(C_f) = a_s \frac{\sigma_w}{b_s + \sigma_w}, \quad (18)$$

where a_s and b_s are fitting parameters, and subscript s is to signify that these are the fitting parameters for the SLP model. Parameter a_s is considered linearly dependent on the ionic mobility of the counterions in the Stern layer (β_s), which for simplicity is considered to be independent of salinity. Equation 18 can be directly compared with equation 12.

[Skold et al. \(2011\)](#) recently presented an extension of this model that accounts for an additional polarization mechanism beyond the Stern layer polarization that these authors attribute to hopping of the protons on the mineral surface. This extension to the SLP model is used to explain the pH dependence of the imaginary conductivity observed for silica sands.

To account for this effect, [Skold et al. \(2011\)](#) include an additional contribution to the specific surface conductivity,

$$\Sigma_s = \Sigma_s(C_f) + \Sigma_s(H^+), \quad (19)$$

where the second term represents the possibility of mobile protons. [Skold et al. \(2011\)](#) propose the following form for the proton term,

$$\Sigma_s(H^+) = e \beta(H^+) \Gamma_{>\text{SiOH}_2^+}, \quad (20)$$

where $\Gamma_{>\text{SiOH}_2^+}$ is the concentration of surface sites for which protons are mobile and $\beta(H^+)$ represents an effective mobility of the proton along the mineral surface. The addition of this proton contribution implies a modification to equation 18 of the form,

$$\sigma''(\sigma_w, H^+) \propto \Sigma^s(C_f, H^+) = c_s + \left(a_s \frac{\sigma_w}{b_s + \sigma_w} \right), \quad (21)$$

where c_s is a third model parameter linearly related to $\Gamma_{>\text{SiOH}_2^+}$ and $\beta(H^+)$, both of which are considered independent of salinity. [Skold et al. \(2011\)](#) state that this proton hopping mechanism can dominate at very low pH, but may not be negligible at high pH.

METHODS AND DATA SETS

Our database is made up of 67 samples with σ'' recorded as a function of salinity from multiple sources ([Vinegar and Waxman, 1984](#); [Flath, 1989](#); [Lesmes and Frye, 2001](#); [Slater and Lesmes, 2002](#); [Slater and Glaser, 2003](#); [Comas and Slater, 2004](#); [Breede, 2006](#); [Kruschwitz, 2008](#); [Schröder, 2008](#); [Revil and Skold, 2011](#)). Table 1 summarizes the database, and includes the σ_w range investigated, the selected measurement frequency, the source of the data set and a sample description. The database includes fifteen previously unreported unconsolidated samples (# 43–57). Samples were measured at a pH close to neutral.

All measurements reported in Table 1 were obtained by measuring the impedance magnitude and phase shift of the voltage waveform recorded across the sample relative to the current waveform recorded on a reference resistor, the source typically being a sine signal (e.g., [Slater and Lesmes, 2002](#)). All measurements reported here were recorded using a four-electrode device, whereby separate electrode pairs were used to inject the current into the sample and record the potential waveform across the sample. Measured magnitude and phase were converted into a measured complex conductivity (σ^*) using the geometric factor defining the measurement geometry and the resulting current flow path in the test device.

Complex conductivity measurements can exhibit significant frequency dependence. It may then be necessary to model this frequency dependence to extract parameters that quantify the integrated polarization response over the measured frequency range. For example, a normalized chargeability can be defined from fitting either Cole-Cole ([Slater et al., 2005](#)), generalized Cole-Cole ([Scott and Barker, 2005](#)), or Debye decomposition ([Nordsieck and Weller, 2008](#)) models of measured spectra. Many of the data sets we use, which go back up to 28 years (Table 1), only report the dependence of the polarization (σ'') on salinity for a single frequency. To examine the largest database possible, we therefore focus on analyzing just a single frequency measurement. Assuming that the frequency dependence is not very strong, we can expect normalized chargeability and σ_w to show a similar dependence as σ'' and σ_w ([Weller et al., 2011](#)). However, the relationship between σ'' and σ_w is likely to show some degree of frequency dependence. This

Table 1. Summary of the 67 samples used in this study, showing source for each sample, description of sample, key measurement parameters, and findings (b is power law exponent of fit of σ'' versus σ_w below the polarization maximum).

	Sample	Salinity range (mS/m)	Frequency (Hz)	b	Asymptote reached?	Polarization peak? (σ_w mS/m)	Description
	1. Sandstones						
	Breede (2006)						
1	BK	7–2200	0.05	0.2	Y	Y(1200)	Helsby formation
2	BS4	7–2200	0.05	0.34	Y	Y(1100)	Kiddminster formation
3	BR5	7–2200	0.05	0.27	N	N	Helsby formation
	Schröder (2008)						
4	B49H	25–630	0.05	0.25	N	N	Bahariya formation
5	B49V	25–630	0.05	0.25	N	N	Bahariya formation
6	B4H	25–630	0.05	0.34	N	N	Bahariya formation
7	B4V	25–630	0.05	0.3	N	N	Bahariya formation
8	ES14	20–3000	0.09	0.4	N	N	Elbe sandstone
9	OK4	20–2000	0.05	0.32	Y	Y(320)	Oberkirchen sandstone
10	BU1	20–3000	0.05	0.3	N	N	Bunter sandstone
	Breede (2006); also shown in Weller et al. (2011)						
11	GR	24–2100	0.05	0.45	Y	Y(1200)	Glauconitic sandstone
12	BU12	96–2200	2.93	0.6	N	N	Bunter sandstone
13	BU3	24–2300	0.05	0.38	N	N	silty Bunter sandstone
	Kruschwitz (2008)						
14	B15	30–1000	0.15	0.59	N	N	Baumberg sandstone
15	C15	30–1000	0.1	0.46	N	N	Cotta sandstone
	Lesmes and Frye (2001)						
16	B_LF	20–8000	5	0.26	Y	Y(1000)	Berea sandstone
	Börner (1992)						
17	E10	3–5100	1	0.38	Y	Y(900)	Elbe sandstone
18	E12	3–5100	1	0.4	N	N	Elbe sandstone
19	B2	3–5300	1	0.25	Y	Y(900)	Bentheim sandstone
20	F3	2–1600	1	0.19	Y	Y(1600)	Fontainebleau sandstone
	Flath (1989)						
21	H18H	20–10000	1	0.34	Y	Y(1600)	Cheshire Basin
22	HCF120H	20–10000	1	0.38	Y	Y (2100)	Cheshire Basin
23	HCF236V	20–10000	1	0.31	Y	Y(2900)	Cheshire Basin
	Vinegar and Waxman (1984)						
24	3411	100–15000	30	0.21 ³	N	N	shaly sandstones
25	3336A	100–15000	30	0.1 ³	N	N	shaly sandstones
26	3478	100–15000	30	0.17 ³	N	N	shaly sandstones
27	101	100–1500	30	0.23	Y	N	shaly sandstones
28	102	100–8500	30	0.35	Y	Y(1100)	shaly sandstones
29	CZ10	100–15000	30	0.16 ³	N	N	shaly sandstones
30	3833A	100–15000	30	0.2 ³	N	N	shaly sandstones
31	3126B	100–15000	30	0.28 ³	N	N	shaly sandstones
32	3847A	100–8300	30	0.2	Y	Y (2500)	shaly sandstones
33	3823A	100–15000	30	0.32 ³	Y	Y(4600)	shaly sandstones
34	3885B	100–15000	30	0.34 ³	Y	Y(4600)	shaly sandstones
35	3972E	100–15000	30	0.31 ³	N	N	shaly sandstones

Table 1. Summary of the 67 samples used in this study, showing source for each sample, description of sample, key measurement parameters, and findings (b is power law exponent of fit of σ'' versus σ_w below the polarization maximum). (continued)

	Sample	Salinity range (mS/m)	Frequency (Hz)	b	Asymptote reached?	Polarization peak? (σ_w mS/m)	Description
36	3258A	100–15000	30	0.36 ³	Y	Y(4600)	shaly sandstones
37	3891A	100–15000	30	0.35 ³	Y	Y(4600)	shaly sandstones
38	3308A	100–15000	30	0.37 ³	Y	Y(4600)	shaly sandstones
39	3323F	1000–15000	30	n.d.	Y	Y(4600)	shaly sandstones
40	3324A	1000–15000	30	n.d.	Y	Y(4600)	shaly sandstones
41	3323E	1000–15000	30	n.d.	Y	Y(4600)	shaly sandstones
42	3324B	1000–15000	30	n.d.	Y	Y(4600)	shaly sandstones
2. Unconsolidated sediments							
43	B3_2	3–300	1	0.3	N	N	sands and gravels
44	B2_4	3–300	1	0.45	N	N	sands and gravels
45	B6_7	3–300	1	0.26	N	N	sands and gravels
46	C5_5	3–300	1	0.33	N	N	sands and gravels
47	B1	3–300	1	0.28	N	N	sands and gravels
48	C1_5	3–300	1	0.28	N	N	sands and gravels
49	C2_7	3–300	1	0.33	N	N	sands and gravels
50	C5	3–300	1	0.23	N	N	sands and gravels
51	C3_5	3–300	1	0.23	N	N	sands and gravels
52	B4_7	3–300	1	0.34	N	N	sands and gravels
53	C3_3	3–300	1	0.25	N	N	sands and gravels
54	C4_2	3–300	1	0.28	N	N	sands and gravels
55	C6197	80–400	1	0.43	N	N	sand and gravel flood deposits
56	C6189	10–120	1	0.4	N	N	sand and gravel flood deposits
57	C6208	10–120	1	0.3	N	N	sand and gravel flood deposits
Slater and Lesmes (2002)							
58	G1	2–200	1	0.36	Y	Y(30)	Maine sand and gravel
Comas and Slater (2004)							
59	S1	2–900	1	0.55	N	N	Peat soils (organics)
60	C1	2–1600	1	0.51	N	N	Peat soils (organics)
61	C9	2–1600	1	0.51	N	N	Peat soils (organics)
Slater and Glaser (2003)							
62	N2: 27–35	50–6600	1	0.42	Y	Y(850)	Sand/silt floodplain sediments
63	N2: 19–25	57–3500	1	0.64	Y	Y(140)	Sand/silt floodplain sediments
64	N1: 26.5–35	57–3800	1	0.35	Y	Y(260)	Sand/silt floodplain sediments
65	N1: 51–53	57–3300	1	0.37	Y	Y(3300)	Sand/silt floodplain sediments
Revil and Skold (2011)							
66	SB	3–200	0.1	0.58	N	N	Artificial sand
67	U#30	10–1000	0.1	0.21	N	N	Artificial sand

³Regression is only two points below 1000 ms/m.

should be recognized when comparing the data sets in Table 1, because the measurement frequency varies from a minimum of 0.05 to a maximum of 30 Hz.

The analysis of each data set first involved fitting a simple power law function to the observed general increase in σ'' with σ_w at the lower salinities,

$$\sigma'' = a(\sigma_w)^b, \quad (22)$$

where a and b are fitting parameters. Although equation 22 has no physical basis, it does provide an assessment of the first-order variations in the strength of the salinity dependence of σ'' over the entire sample database. Many of the data sets are characterized by a high salinity asymptote reported in previous measurements and predicted by the DLP and SLP models. Therefore, equation 22 was only fit to all available data below 1000 mS/m (representing potable groundwater) unless the asymptote occurred at lower

salinities, as observed for some unconsolidated samples. In the latter case, equation 22 was only fit to the σ_w measurements at salinities lower than the asymptote. In all except 12 samples, a minimum of three data points were available to define b . The exceptions occur for 12 of the 19 data sets used here that were reported by Vinegar and Waxman (1984) and highlighted (superscript a) in Table 1. Vinegar and Waxman (1984) were primarily interested in the characterization of briny oil reservoirs, and therefore focused on a higher salinity range than the other data sets reported in Table 1. Consequently, most of their measurements capture the σ'' dependence close to/on the asymptote, with limited coverage at the lower salinities that are more representative of potable groundwater. Values of the power law exponent b are presented in Table 1 and discussed later.

The remaining data analysis involved evaluating the fits of the DLP and SLP salinity dependence described in equations 12 and 21 to the σ'' data recorded across the σ_w range prior to the asymptote for each sample. Each data set also was evaluated for evidence of a σ'' peak and decrease in polarization at highest salinity.

Finally, to investigate evidence of the polarization decrease at high σ_w further, we took one sandstone sample (#11, GR from Breede, 2006) with measurements up to 2200 mS/m and extended these measurements to $\sim 10,000$ mS/m by acquiring data with three additional saturating electrolytes ($\sigma_w = 2800, 4500$, and 9000 mS/m). Complex conductivity measurements were acquired for each saturating electrolyte over a frequency range from 0.003 to 100 Hz using two commercially available SIP instruments, the SIP Fuchs (Radic Research) and the SIP Jülich described in Zimmermann et al. (2008). This comparison of instruments was performed to investigate whether the polarization decrease could be an artifact of measurement errors at high σ_w . Equation 1 shows that the phase measurement used to quantify the polarization response is inversely proportional to σ' . Therefore, φ decreases with increasing σ_w and

could approach the accuracy of the instrumentation. Whereas modern commercially produced SIP instruments have a high φ accuracy (e.g., ~ 0.1 mrad, Zimmermann et al., 2008), the accuracy of the instruments used to acquire the historic data sets reported in this paper is somewhat uncertain.

RESULTS

We organize the results by first presenting the fit of the data sets to salinity dependence predicted by the DLP and SLP models, as well as the fit to the simple empirical function described by equation 22. We then proceed to consider the significance of the parameters with respect to their dependence on available petrophysical properties and their predicted values in the case of the SLP model.

Fitting quality of the models

Table 1 summarizes the findings of the analysis performed on all 67 samples. The power law exponent (b) obtained from fitting equation 22 is reported for all samples excluding #39–42 from Vinegar and Waxman (1984). These four samples do not contain at least two data points for the fitting below the asymptote, but are retained in this analysis to support our investigation of the significance of a distinct polarization peak. Data sets exemplifying the power law behavior below 1000 mS/m are shown in Figure 1. Representative sandstone samples from four previous independent studies are plotted in Figure 1a. Representative unconsolidated samples from two previous independent studies and two new databases of our own are plotted in Figure 1b. The sandstone data sets are plotted to higher σ_w and show evidence of the plateau or maximum being reached above 1000 mS/m.

Considering the entire database ($n = 63$), the exponents vary from a minimum of 0.1 to a maximum of 0.64, with a mean (μ)

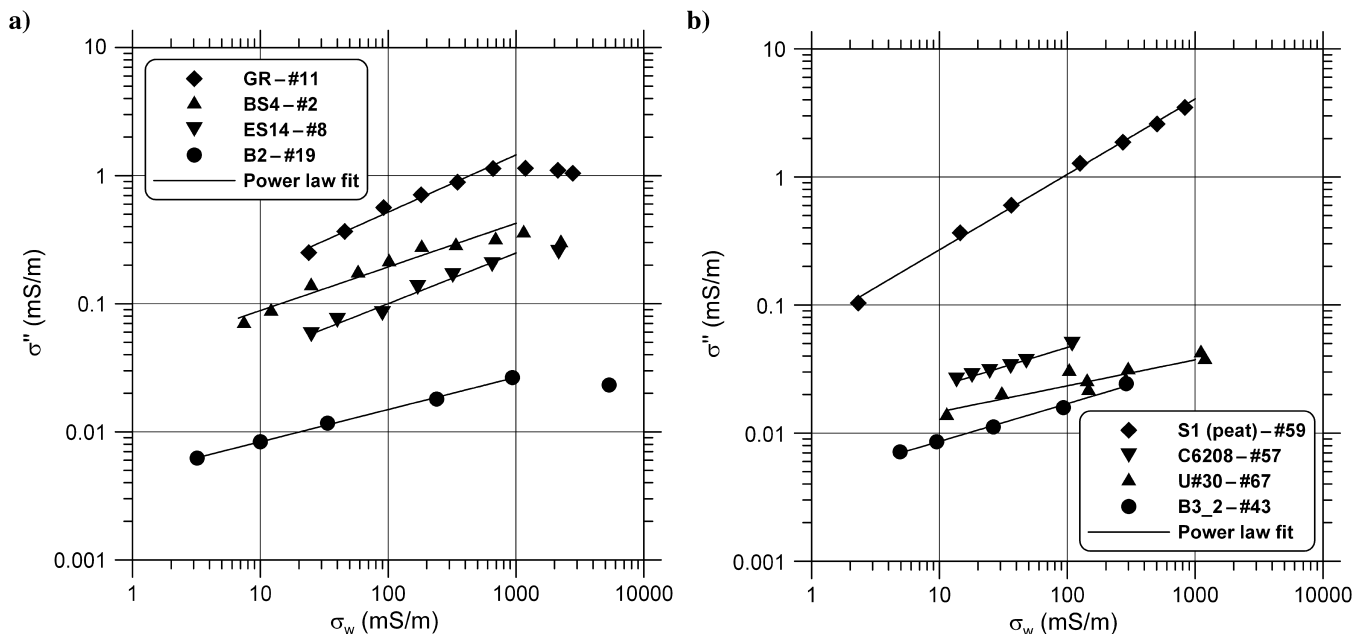


Figure 1. Example data sets showing the fit of the power law dependence of σ'' on σ_w below the asymptote (where present): (a) sandstone samples; (b) unconsolidated samples. Sample descriptions and power law exponents (b) are given in Table 1.

of 0.34 and a standard deviation of (σ_d) of 0.11. Excluding the high-salinity (Vinegar and Waxman, 1984) data set that contains more data close to the polarization asymptote ($n = 48$), the minimum increases to 0.19 and μ increases to 0.36, whereas the maximum and σ_d are unchanged. Considering the sandstone samples alone, the minimum is 0.1, the maximum is 0.59, $\mu = 0.31$, and $\sigma_d = 0.11$ ($n = 38$). Excluding the Vinegar and Waxman (1984) data set, this minimum increases to 0.19, μ increases to 0.35, and $\sigma_d = 0.11$ ($n = 23$). Considering the unconsolidated samples alone, the minimum is 0.21, the maximum is 0.64, and μ increases to 0.37 and $\sigma_d = 0.12$ ($n = 25$).

Twenty-three sandstone samples and five unconsolidated samples spanning nine independent data sets were found to display strong evidence of a high-salinity asymptote in the $\sigma''(\sigma_w)$ data. Figure 1a illustrates examples of this asymptotic behavior at $\sigma_w > 1000$ mS/m for sandstone samples from four independent data sets. This asymptote is predicted by the DLP (equation 12) and SLP (equation 21) models. However, twenty seven of these samples (again spanning nine independent data sets) exhibit some evidence of a polarization peak, with a decrease in polarization at high salinities that is not predicted by either model. Examples of this polarization peak are shown in Figure 2 for sandstone samples from three independent data sets (Figure 2a) and unconsolidated samples from one data set (Figure 2b). Table 1 indicates the salinity at which this peak occurs ($\sigma_{w[p]}$). For the 27 sandstone samples, the minimum $\sigma_{w[p]}$ is 320 mS/m, the maximum is 4600 mS/m, $\mu = 2719$ mS/m and $\sigma_d = 1690$ mS/m. These statistics are influenced strongly by the Vinegar and Waxman (1984) data set, containing nine samples where the peak is at 4600 mS/m, being the maximum within the identified range. For the five unconsolidated samples, the minimum $\sigma_{w[p]}$ is 30 mS/m, the maximum is 3200 mS/m, $\mu = 916$ mS/m and $\sigma_d = 1370$ mS/m. This analysis demonstrates that this polarization peak occurs at significantly lower salinity for the unconsolidated sediments.

Most of the data sets could be well-fit by the salinity dependence of the DLP model below the asymptote as described by equation 12. Figure 3 shows selected examples of the fitting quality to (a) sandstone samples, and (b) unconsolidated samples. The samples were selected to represent results obtained from independent studies and over different salinity ranges. The data shown above the polarization peak for sample # 21 are excluded from the fitting as the high σ_w polarization decrease is not predicted by the model. In total, the salinity dependence of the DLP model was found to describe 56 of the 67 samples well.

The fit of the SLP model to the available data sets was found to be critically dependent on the inclusion of the additive constant c_s recently proposed by Skold et al. (2011) to describe the additional contribution of the protons to the polarization. We examined the fit of the data sets to two forms of the model (a) with $c_s = 0$ as proposed by Revil and Skold (2011) in their studies of the salinity dependence of the induced polarization response of sandstones, and (b) with $c_s > 0$ as proposed by Skold et al. (2011) to account for the protons. In the case of $c_s = 0$ (equation 18), the dependence of σ'' on σ_w predicted by the SLP model was found only to provide a better fit than the DLP model (equation 12) for seven samples in our database (this included only one unconsolidated sample). Under these conditions, the deviation of the SLP model from the experimental data results from the fact that the slope of the σ'' dependence on σ_w approaches one at low salinity. However, the inclusion of the additional polarization contribution represented by c_s (equation 21) resulted in an excellent fitting to most data sets (excluding any data points suggesting a decrease in σ'' at high salinity). Figure 4 shows example fits to selected sandstone samples (Figure 4a) and unconsolidated samples (Figure 4b). Samples shown in Figure 4 were selected to represent results obtained from independent studies and to capture evidence of the asymptote (in log-log conductivity space) predicted when $c_s > 0$. The fits for $c_s = 0$ are shown by the dashed lines and the fits for $c_s > 0$ are

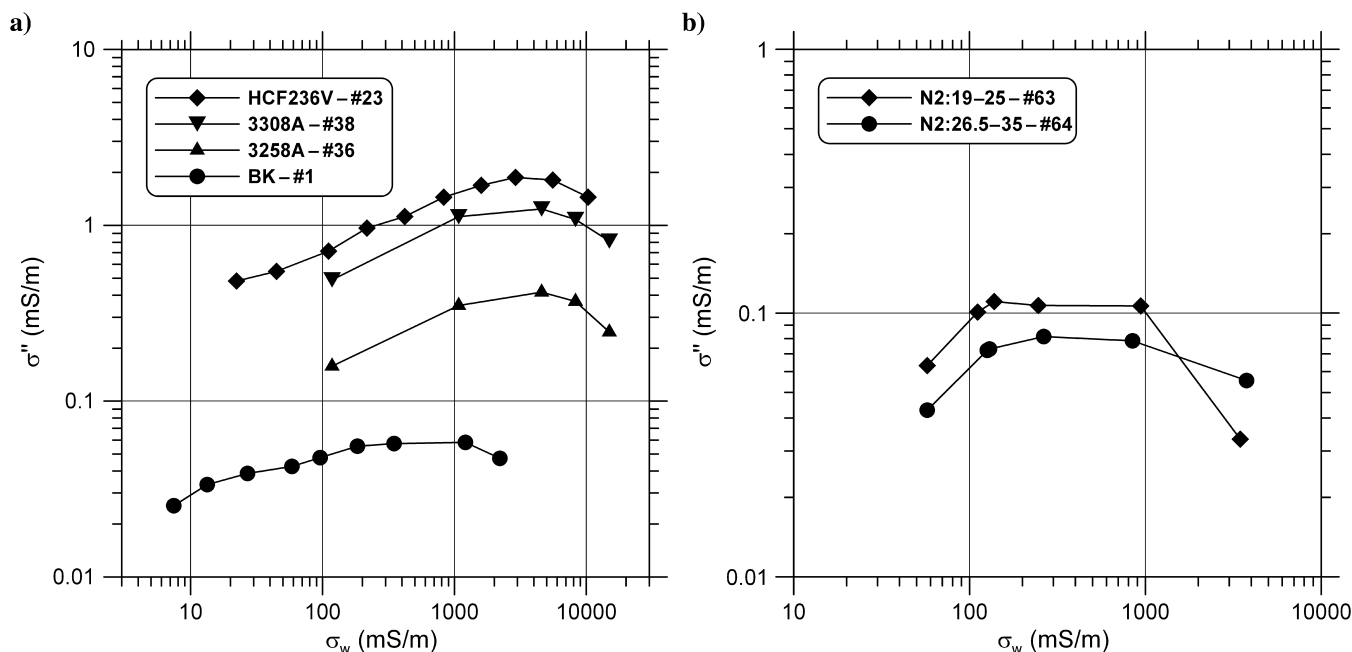


Figure 2. Example data sets showing evidence for a decrease in σ'' at high salinity; (a) sandstone samples, (b) unconsolidated samples.

shown by the solid lines. Two unconsolidated samples capture evidence for the low salinity asymptote predicted by the SLP model with $c_s > 0$ (samples # 44 and # 61). Sample # 61, an organic peat sample with measurements over a wide salinity range, provides strong support for an additive contribution not predicted by either equation 12 or 18. We note, however, that these peat samples are not representative of the clean sands for which the SLP is derived.

Four of the samples from the Vinegar and Waxman (1984) database could not be fit to either the SLP or DLP model because data points prior to the polarization peak are missing. These are samples that show a pronounced peak in the polarization and contain multiple data points indicating a drop in σ'' at high salinity. As noted previously, this high-salinity decrease in polarization is not predicted by either the DLP or SLP models. Results of our extended

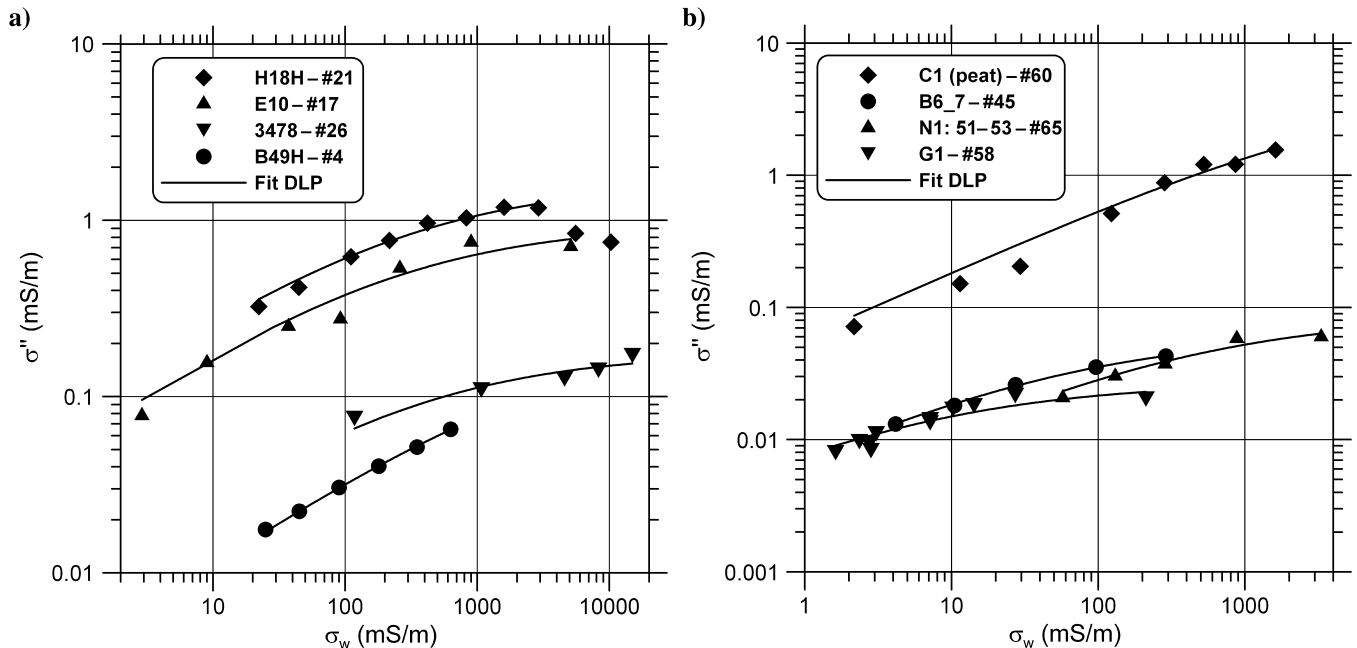


Figure 3. Example data sets showing the fit of the DLP model to experimental data sets; (a) selected sandstone samples, (b) selected unconsolidated sediments.

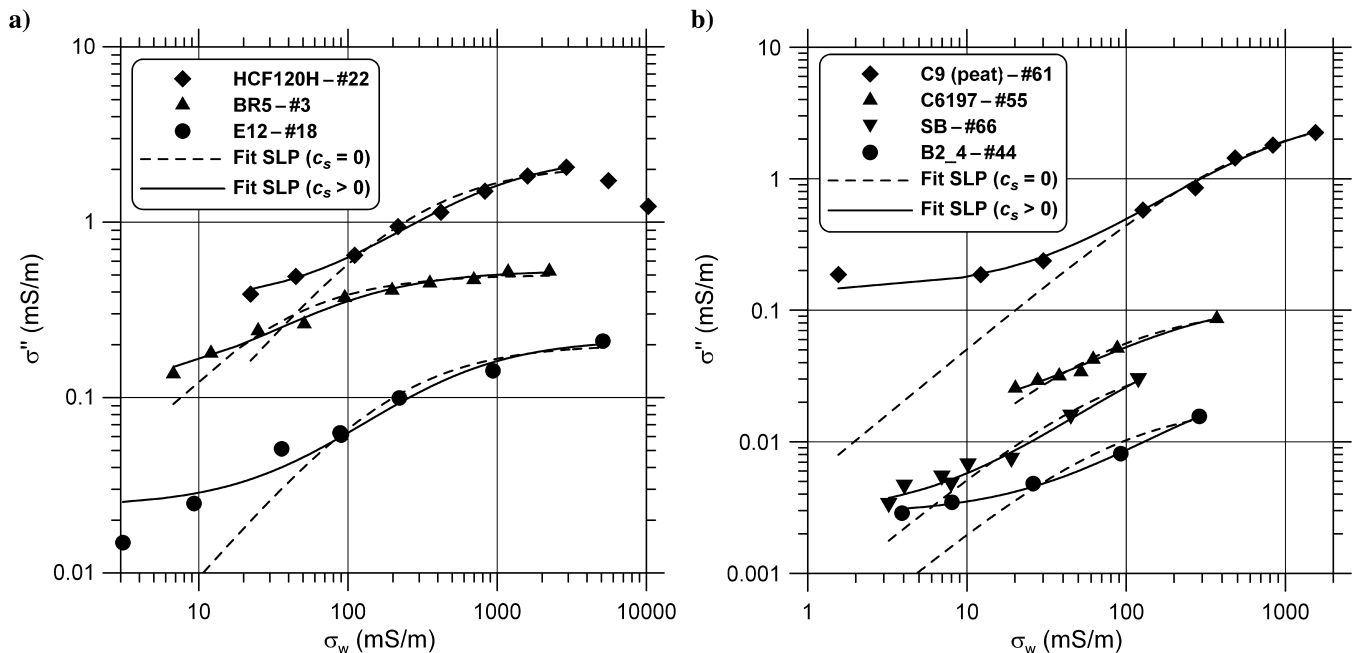


Figure 4. Example data sets showing the fit of the SLP model to experimental data sets; (a) selected sandstone samples, (b) selected unconsolidated sediments. The fits when the contribution of the protons is neglected ($c_s = 0$, e.g., Revil and Skold, 2011) and when the additional contribution is considered ($c_s > 0$, e.g., Skold et al., 2011) are shown for comparison.

high-salinity measurements using two high-accuracy, commercially produced complex resistivity instruments are shown in Figure 5. Figure 5a compares the full phase spectra for the three high-salinity measurements acquired with the two instruments. The response from the two instruments is near identical over the entire frequency range of the measurements. Figure 5b shows the extension of the measured σ'' dependence on σ_w to high salinities beyond the asymptote originally recorded in this data set. These measurements support the existence of a polarization peak and a high-salinity decrease in σ'' as has been previously recorded for 27 samples from nine independent data sets (Table 1).

Physical significance of the fitting parameters

We next consider the physical significance of the fitting parameters derived from the DLP and SLP models. We use 20 sandstone samples that have estimates of S_{por} and F , along with complex conductivity measurements at more than five salinities. The petrophysical properties and fitting parameters for the DLP and SLP models are shown in Table 2. We first consider the DLP model and the significance of the model parameters shown in equation 12. Data for 19 of the 20 samples reported in Table 2 were described well by the DLP model. Because no reasonable fitting parameters of sample B15 were achieved, this sample was excluded from the interpretation.

Although the DLP model is not parameterized sufficiently to assign expected values to the fitting parameters, equation 12 does provide information on the expected dependence of the model parameters on physical properties. Parameter a_d , which varies over almost three orders of magnitude (4.2×10^{-3} to 2.2 mS/m), describes the asymptotic value of σ'' at high salinity. This parameter should be strongly related to S_{por}/F (equation 10). Indeed, the variation of the normalized parameter $a_d F/S_{\text{por}}$ is much reduced to a little over one order of magnitude (202–3961 pS). However, the normalized parameter a_d/S_{por} , according to the concept of specific

polarizability (Weller et al., 2011), results in a further reduced range from 13–132 pS (with an average value 42 pS), thereby indicating a near-linear relation between a_d and S_{por} . The power law fit describing this relation is given by

$$a_d = 0.0304 S_{\text{por}}^{1.04}, \quad (23)$$

(a_d in mS/m and S_{por} in $1/\mu\text{m}$) with a coefficient of determination $R^2 = 0.822$ (Figure 6a).

The parameter b_d is related to the salinity dependence of imaginary conductivity. Lower values of b_d indicate asymptotic behavior of $\sigma''(\sigma_w)$ being reached at lower salinity. The numerical value of b_d varies between 4 and 134 with a mean value of 25. A slight increase in b_d with increasing S_{por} is observed. A power law fit of b_d to S_{por} results in an exponent of 0.3, but the coefficient of determination is only 0.197.

We next turn to the significance of the fitting parameters (equation 21) with respect to the SLP model. Unlike the DLP model, the expected values of these parameters are defined by the theory (Revil and Skold, 2011; Skold et al., 2011). We found that all 20 samples reported in Table 2 could be fit by equation 21. However, two samples (samples # 9 and # 16 in Table 2) gave a negative value of c_s , which is physically unreasonable but can be explained by the fact that c_s is only well-resolved when the low salinity asymptote is visible in the fitted curves (not the case in these samples). These two samples are therefore excluded from this analysis.

We first note that a_s should be an analog of a_d in the DLP model. Parameter a_s , which varies by almost three orders of magnitude (0.00272 to 1.21 mS/m), describes the asymptotic value of σ'' at high salinity. This parameter should be strongly related to S_{por}/F (equation 16). Considering equations 13, 14, 15, and 17, the normalized parameter $a_s F/S_{\text{por}}$ can be expressed by

$$\frac{a_s F}{S_{\text{por}}} = \frac{1}{12} \beta_{\text{Na}} e \Gamma_s^0. \quad (24)$$

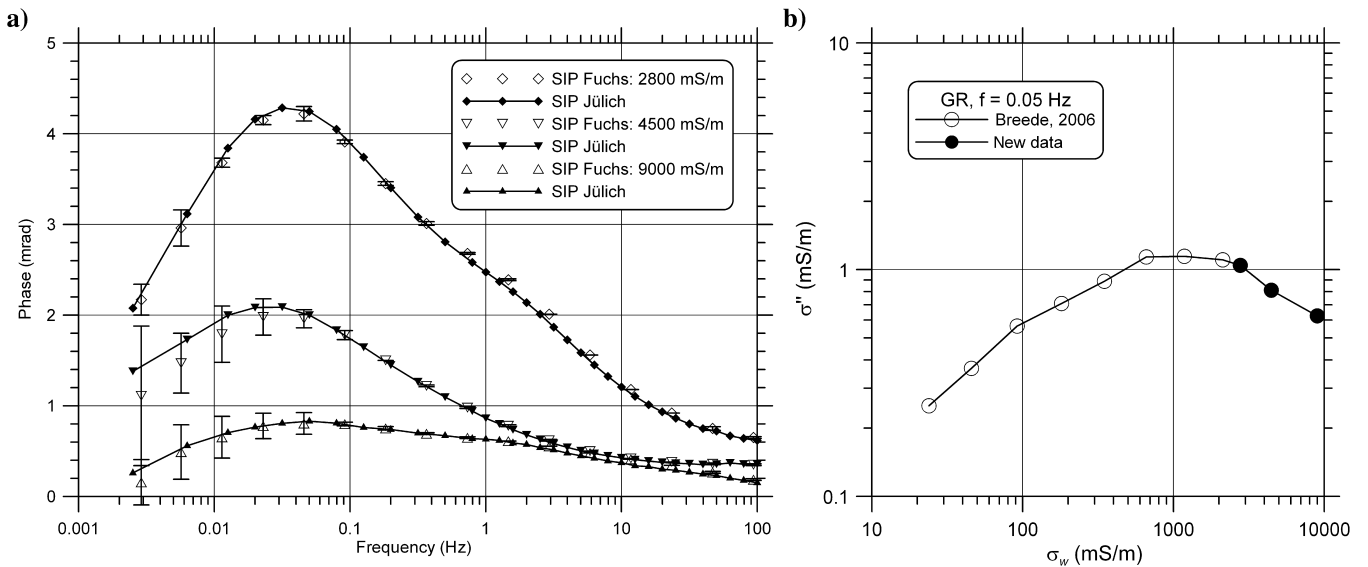


Figure 5. Results for sample GR #11. (a) Phase spectra recorded at high salinity with two different instruments (phase is shown positive for conductivity space and individual error bars are plotted for SIP Fuchs instrument); (b) σ'' at 0.05 Hz versus σ_w showing the extension of the original data to high salinity and the clear evidence for the high-salinity polarization decrease.

Assuming an average site density of 5 sites nm⁻² and a mobility $\beta_{\text{Na}} = 5.19 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ (adopted from [Revil and Skold, 2011](#)), this normalized parameter should be on the order of 3500 pS. Our fitting predicts $a_s F / S_{\text{por}}$ between 128 and 2544 pS with an average value of 539 pS. This difference might indicate a lower site density and ionic mobility in our samples than what is assumed by the model. The normalization of a_s to S_{por} according to the concept of specific polarizability (c_p) ([Weller et al., 2011](#)) results in a further reduced range from 7.7 to 48 pS (average value 20 pS) and confirms a nearly linear relation between a_s and S_{por} . The power law fit describing this relation is given by

$$a_s = 0.0158 S_{\text{por}}^{1.04} \quad (25)$$

(a_s in mS/m and S_{por} in 1/ μm) with $R^2 = 0.890$ (see Figure 6b). [Revil \(2012, equation 48\)](#) gives an explicit relation for a_s developed for shaly sands. Taking average values for all parameters and ignoring the contribution of protons, he calculates a value of 12.3 pS for the specific polarizability (Equation 5), being close to our average of 20 pS.

Similar to b_d , from the DLP model, parameter b_s is related to the salinity dependence of imaginary conductivity. Lower values of b_s indicate asymptotic behavior of $\sigma''(\sigma_w)$ being reached at lower salinity. The numerical value of b_s estimated from the fitting varies between 37 and 747 with a geometric mean value of 190. Considering the approximation,

$$\sigma_w [\text{mS/m}] \cong 10000 \times C_f [\text{mol/l}], \quad (26)$$

the dimensionless parameter b_s can be expressed by

$$b_s = \frac{10000 \times 10^{-\text{pH}}}{K_{\text{Na}}}, \quad (27)$$

and depends on the pH and the sorption coefficient of sodium. Assuming pH = 6 and $K_{\text{Na}} = 10^{-3.25}$ (adopted from [Revil and Skold \[2011\] Figure 8](#)), we get $b_s = 18$. This value is approximately a factor of 10 lower than the resulting geometric mean; this could indicate a lower average pH or K_{Na} for our sample database than the assumed values.

Table 2. Petrophysical parameters for 20 sandstone samples from the compiled database and fitting parameters for the DLP and SLP models: All samples listed in this table had a minimum of five data points describing the dependence of σ'' on σ_w .

Sample		Porosity	F	$S_{\text{por}} (\mu\text{m}^{-1})$	DLP model		SLP model		
					a_d (mS m $^{-1}$)	b_d	a_s (mS m $^{-1}$)	b_s	c_s (mS m $^{-1}$)
Breede (2006)									
1	BK	0.160	16.6	5.12	0.678	4.02	0.039	37.26	0.021
2	BS4	0.166	17.6	28	0.500	13.82	0.308	86.25	0.052
3	BR5	0.243	8.2	15	0.637	8.47	0.415	72.83	0.114
Schröder (2008)									
4	B49H	0.160	26.8	3.29	0.058	10.99	0.032	125.20	0.013
5	B49V	0.160	31.0	3.54	0.061	12.31	0.035	224.70	0.016
6	B4H	0.120	44.9	8.93	0.224	30.68	0.128	484.40	0.030
7	B4V	0.130	59.6	8.29	0.129	19.40	0.071	251.20	0.022
8	ES14	0.180	18.7	5.2	0.424	26.98	0.251	218.70	0.031
9	OK4	0.180	22.1	8.23	0.236	12.62			
10	BU1	0.140	33.8	25.4	0.852	12.31	0.552	100.00	0.113
Breede (2006); also shown in Weller et al. (2011)									
11	GR	0.246	9.1	31.3	1.775	20.45	1.171	130.80	0.066
12	BU12	0.181	18.0	10.5	1.106	134.71	0.350	581.30	0.002
13	BU3	0.089	66.7	46.2	1.540	60.00	0.738	746.80	0.121
Kruschwitz (2008)									
14	B15	0.191	12.6	69.17			1.211	582.20	0.066
15	C15	0.217	11.6	50.8	2.219	45.60	0.926	142.90	0.029
Lesmes and Frye (2001)									
16	B_LF	0.180	15.9	8.3	0.126	7.41			
Börner (1992)									
17	E10	0.224	17.0	41.6	0.949	15.19	0.694	156.10	0.088
18	E12	0.259	11.0	5.5	0.311	34.97	0.191	389.70	0.024
19	B2	0.194	15.6	2.15	0.028	7.08	0.019	97.10	0.006
20	F3	0.068	118.0	0.126	0.004	6.38	0.003	192.50	0.001

We finally consider parameter c_s introduced by Skold et al. (2011) to describe a contribution of the protons to the polarization. In our fitting, this additive constant varies between 1.34 and 121 $\mu\text{S}/\text{m}$. Similar to the parameter a_s , a normalized parameter $c_s F/S_{\text{por}}$ can be explicitly formulated:

$$\frac{c_s F}{S_{\text{por}}} = \frac{1}{12} \beta(H^+) e \Gamma_{>\text{SiOH}_2^+} = \frac{1}{12} \Sigma^S(H^+). \quad (28)$$

Assuming $\Sigma^S(H^+) \approx 2$ nS as shown in Skold et al. (2011) (see their Figure 2a), the value of the normalized parameter $c_s F/S_{\text{por}}$ is expected to be on the order of 160 pS, which is very close to the average value of 143 pS that is determined by the fitting procedure. However, this normalized parameter varies over a wide range from 3.2 to 1250 pS; this is likely attributable to the poor resolution of parameter c_s for many samples due to limited data points defining the low salinity asymptote (in double logarithmic space) predicted by equation 21.

DISCUSSION

This analysis of an extensive database of multisalinity, complex conductivity data sets demonstrates that σ'' universally follows a simple positive power law dependence on σ_w at low salinities representative of natural groundwater. Such an increase in polarization with increasing σ_w is predicted by the DLP and SLP models to result from an increase in surface charge density as σ_w increases. In most of the samples reported, this power law dependence is valid below $\sigma_w = 1000$ mS/m. A minority of unconsolidated samples display a polarization asymptote at σ_w significantly less than 1000 mS/m. In this case, the power law dependence only is observed for salinities lower than the asymptote. The mean value of the exponent is 0.34 for the entire database, 0.31 for the sandstone samples, and 0.37 for the unconsolidated samples. The observed dependence of σ'' on σ_w is weak when compared to the

expected linear dependence of σ' on σ_w (equation 3). Such a relatively weak dependence on σ_w , coupled with the strong linear dependence of σ'' on S_{por} (Weller et al., 2010b), highlights the value of complex conductivity measurements in lithologic discrimination and permeability estimation.

Although the empirical fitting and the fitting constants of equation 22 lack physical meaning, this relation does provide a first-order expected dependence of σ'' on σ_w that could be used to compensate for changes in σ'' driven by σ_w variation. Empirical laws frequently are used to constrain field geophysical measurements in the absence of sufficient information on physicochemical properties to populate the model space of physics-based models. Examples of such laws are Archie's law (Archie, 1942) discussed previously and the widely used Topp equation (Topp et al., 1980), a third-order polynomial for estimating moisture content in soils from dielectric permittivity measurements. We have shown here that the empirical power law relation fits the widest available range of sandstone and unconsolidated sediment materials with a relatively narrow range in the power law exponent. This empirical relation could be used to improve complex conductivity derived estimates of S_{por} and permeability for cases where the salinity varies between samples or between field locations. For example, the specific polarizability defined by Weller et al. (2010b) could be defined as a function of salinity.

The salinity dependence predicted by the DLP model mostly satisfies the observation at low salinities quite well for sandstone and unconsolidated sediment samples. The DLP model is based on a relatively simple electrical double layer where the polarization of the Stern layer is considered salinity independent. The DLP model assumes that the salinity dependence arises from an increase in capacity of the diffuse part of the EDL resulting from a decrease in the diffuse layer thickness accompanying an increasing ionic concentration. The mean of the power law exponents is 0.34, and therefore lower than the approximate value of 0.5 (Kulenkampff and

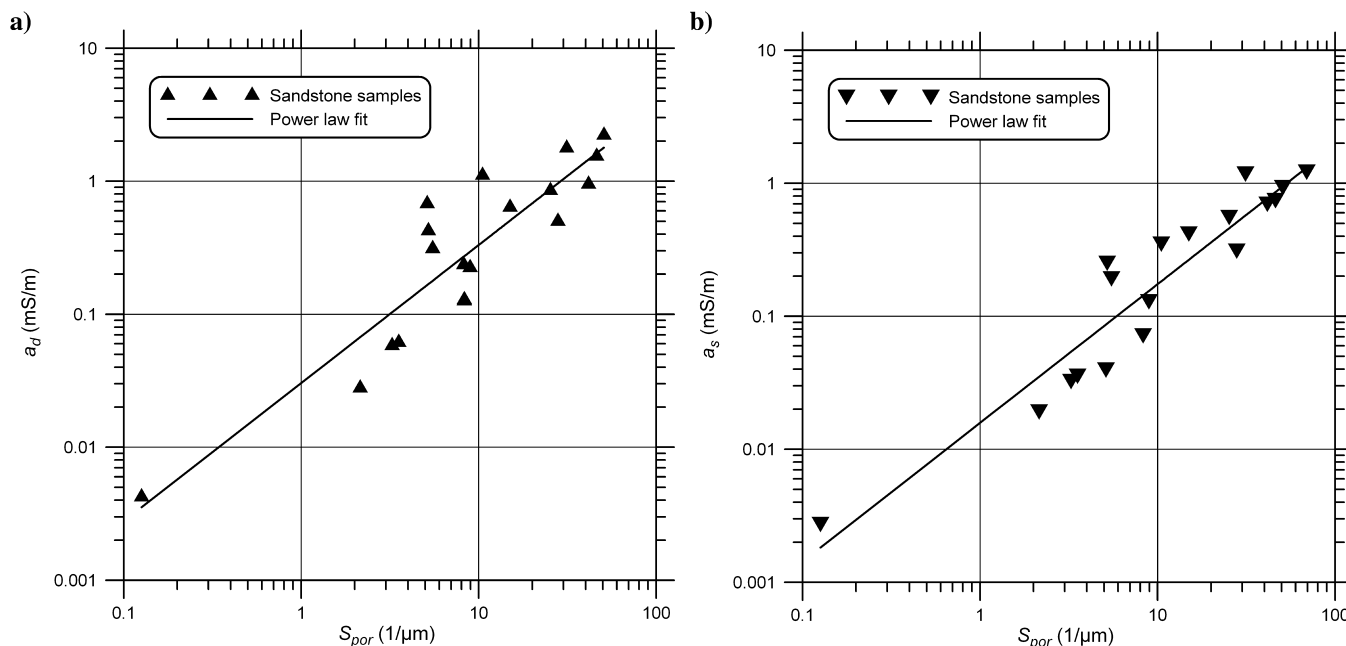


Figure 6. The correlation between model parameters and S_{por} : (a) a_d versus S_{por} ($R^2 = 0.8822$); (b) a_s versus S_{por} ($R^2 = 0.890$).

Schopper, 1988), predicted by the DLP model and previously observed based on the three sandstone samples described in Weller et al. (2011). The value of 0.5 represents the maximum of the slope at sufficiently low σ_w to be well away from the asymptote. The lower average value found here can be interpreted to result from the asymptotic character of the curves and the fact that this curvature is in part captured in many of the power law fits reported here (Table 1). Clearly, fitting data that in part capture the curvature as the asymptote is approached will lower the b estimate relative to that for data well away from the asymptote. For example, the organic sediments (samples # 59–# 61) were all recorded over a large salinity range that captured low salinities well below the asymptote (an example in Figure 3). These samples are fit with a value of $b = 0.5$ – 0.55 , being very close to the expected value from the DLP model.

The fit of the salinity dependence predicted by the SLP model to the observed data sets was found to be critically dependent on whether the recently proposed additive term representing the polarization of the protons (Skold et al., 2011) was included in the model. The form of the SLP model without this additional contribution (e.g., Revil and Skold, 2011) generally did not provide an adequate fit to most of the available experimental data (e.g., Figure 4). However, the inclusion of the term representing the protons generally resulted in an excellent fitting quality to all available data sets (e.g., Figure 4). The addition of this additive term results in a predicted low-salinity asymptote (in double logarithmic space) and evidence of this behavior is captured in a few of the available data sets that extend to low salinity. The strongest evidence for this behavior predicted by the SLP model was found for sample # 61 (shown in Figure 4b), although other samples shown in Figure 4 (# 22 in Figure 4a and # 44 in Figure 4b) also exhibit evidence of this low-salinity behavior predicted by the SLP model when the protons are considered, but not predicted by the DLP model. One clear finding from our work is that the SLP model must include the addition of the recently proposed proton contribution to adequately represent the available data sets describing the salinity dependence of σ_w . The importance of this additional term may therefore be greater than envisaged by Skold et al. (2011), who suggested it mostly would be significant in low-pH environments. It seems reasonable to assume that the role of the protons also would become more dominant toward low salinity due to the reduction in surface charge density within, and hence polarization of, the Stern layer.

Our analysis of the physical significance of the DLP and SLP models provides support for both model frameworks. Although the DLP model is not parameterized sufficiently to derive expected parameter values, we find that the predicted dependence of parameter a_d , representing the polarization at the high-salinity asymptote, is very well supported by the available petrophysical data. The same was observed for parameter a_s , representing the polarization at the high-salinity asymptote for the SLP model. Parameters a_d and a_s are predicted to be controlled strongly by the ratio S_{por}/F . Interestingly, in both cases, we found that a_d and a_s were much more strongly correlated with S_{por} than S_{por}/F . Although the explanation for this finding is unclear and contrary to both models, the result is consistent with the concept of specific polarizability as defined by Weller et al. (2010b). Further work is needed to investigate why the normalization of S_{por} by F results in a weaker correlation with imaginary conductivity.

The SLP model is a sophisticated model in that it combines a macroscopic conductivity model for the polarization of a porous medium with a triple layer model for the specific surface conductivity of the Stern layer. Model parameter values describing the salinity dependence found by fitting the experimental data sets were largely consistent with the expected values based on the available petrophysical properties and SLP model parameter estimates given by others. The relatively small deviations of our fitting parameter values from those predicted from the theory may reflect different pH and salinity conditions in our experiment relative to assumed pH and salinity conditions for cited values of some of the SLP model parameters not available in our database. The good correspondence between observed and predicted fitting parameter values lends additional support to the SLP model framework.

The asymptotic behavior of σ'' at high σ_w predicted by the DLP and SLP model formulations is observed clearly in the samples making up this extensive database. However, there also is compelling evidence of a polarization peak and a subsequent reduction in σ'' at higher salinities. This decrease is not predicted by either the DLP or SLP models. Careful high-salinity measurements using two modern CR instruments indicate that this high-salinity decrease in σ'' is not an artifact resulting from limited measurement precision of small phase angles at high salinity. Lesmes and Frye (2001) suggested that such a decrease in σ'' at high σ_w would result from a reduction in the mobility of the counterions due to the accumulation of charge in the EDL and greater ion-ion interactions. In contrast, the DLP and SLP formulations assume that the surface ionic mobility of the polarized counterions is constant. Revil and Skold (2011) offer an alternative suggestion, being that such a decrease might result from an additional membrane polarization effect not included in their mechanistic model. Irrespective of the physical explanation, it is clear that $\sigma''(\sigma_w)$ may deviate from both models at high salinities. Although this deviation often may occur at salinities greater than the normal range of interest in groundwater investigations, caution must be exercised in the interpretation of low-surface-area unconsolidated sediments where the peak may occur well below 1000 mS/m (e.g., Figure 2b and samples # 58, # 63, and # 64 in Table 1).

CONCLUSIONS

This analysis of a broad database suggests a well-defined salinity dependence of σ'' that is consistent for sandstone and unconsolidated sediments. The existence of an empirical relation, the IP equivalent of Archie's law, previously has been proposed. The results of this new study can improve this relationship by constraining the likely dependence of the specific polarizability, a parameter defining the linear correlation between σ'' and S_{por} , on fluid conductivity. Based on an extensive data set, we find that the power law dependence of σ'' on σ_w at low salinities (up to the polarization maximum representing an asymptote or peak) is on average 0.34. This value is lower than the maximum of 0.5 that is expected from the DLP model and has been reported in a previous study using a limited data set of only three samples.

The data sets generally can be well fit by the salinity dependence predicted by two existing physical models describing induced polarization. Both models result in fitting model parameters that are closely related to pore volume normalized internal surface (S_{por}) as predicted by both theories. However, our findings clearly show that the popular SLP model must include a recently proposed additional

contribution from proton hopping to reproduce the salinity dependence of σ'' toward low salinity. With this addition, the SLP model faithfully reproduces a wide range of experimental data sets, with fitting parameter values generally consistent with values expected from the SLP model theory. However, numerous samples from multiple databases support the existence of a polarization peak at high salinity, with further salinity increases result in a decrease in σ'' . Although the physical explanation for this decrease is not entirely certain, existing models for describing the complex conductivity response of soils and rocks would require modification to capture this high-salinity behavior.

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